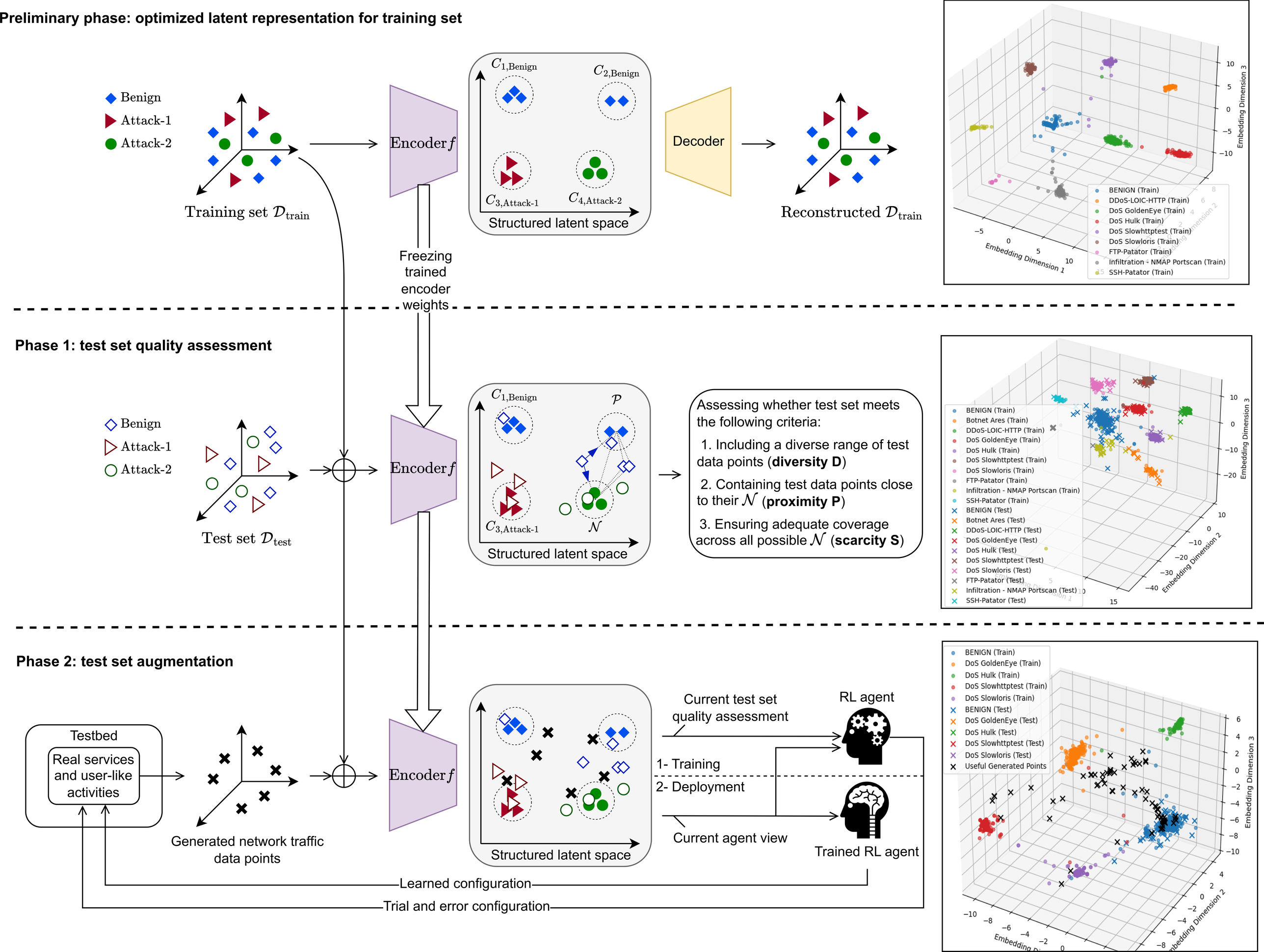
Preliminary phase: optimized latent representation for training set
Benign
Attack-1
Attack-2
Training set $\mathcal{D}_{\text{train}}$
Encoder f
$C_{1,\text{Benign}}$
$C_{2,\text{Benign}}$
$C_{3,\text{Attack-1}}$
$C_{4,\text{Attack-2}}$
Structured latent space
Decoder
Reconstructed $\mathcal{D}_{\text{train}}$
Freezing trained encoder weights
Phase 1: test set quality assessment
Benign
Attack-1
Attack-2
Test set $\mathcal{D}_{\text{test}}$
Encoder f
$C_{1,\text{Benign}}$
$\mathcal{P}$
$C_{3,\text{Attack-1}}$
$\mathcal{N}$
Structured latent space
Assessing whether test set meets the following criteria:
1. Including a diverse range of test data points (**diversity D**)
2. Containing test data points close to their $\mathcal{N}$ (**proximity P**)
3. Ensuring adequate coverage across all possible $\mathcal{N}$ (**scarcity S**)
Phase 2: test set augmentation
Testbed
Real services and user-like activities
Generated network traffic data points
Encoder f
Structured latent space
Current test set quality assessment
RL agent
1- Training
2- Deployment
Current agent view
Trained RL agent
Learned configuration
Trial and error configuration
Embedding Dimension 1
Embedding Dimension 2
Embedding Dimension 3
BENIGN (Train)
DDoS-LOIC-HTTP (Train)
DoS GoldenEye (Train)
DoS Hulk (Train)
DoS Slowhttptest (Train)
DoS Slowloris (Train)
FTP-Patator (Train)
Infiltration - NMAP Portscan (Train)
SSH-Patator (Train)
Botnet Ares (Train)
BENIGN (Test)
Botnet Ares (Test)
DDoS-LOIC-HTTP (Test)
DoS GoldenEye (Test)
DoS Hulk (Test)
DoS Slowhttptest (Test)
DoS Slowloris (Test)
FTP-Patator (Test)
Infiltration - NMAP Portscan (Test)
SSH-Patator (Test)
Useful Generated Points